UF Herbert Wertheim College of Engineering
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# Lecture 2C: Research Methods

EIN 3241: Human Factors and Ergonomics
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# Other ways to describe Research Methods

- Basic vs. Applied
- Quantitative vs. Qualitative
- Descriptive vs. Explanatory
- Field vs. Laboratory

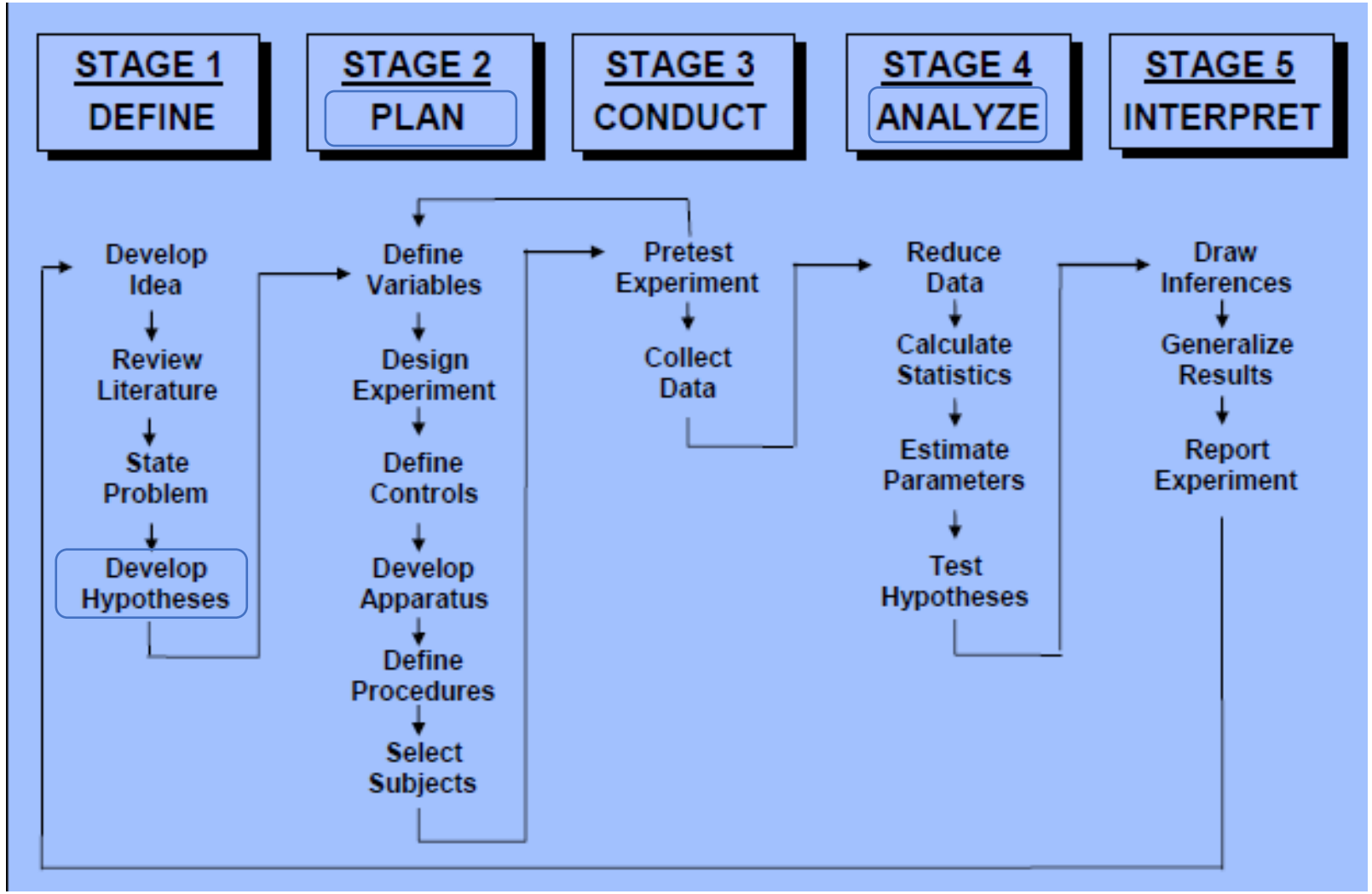
| | | CONSIDERATIONS OF USE? | |
|---|---|---|---|
| | | NO | YES |
| QUEST FOR FUNDAMENTAL UNDERSTANDING? | YES | Pure basic research<br>SYMVATOLOGY<br>THEORETICAL ERGONOMICS | Use-inspired basic research<br>ERGONOMICS DESIGN |
| | NO | | Pure applied research<br>APPLIED ERGONOMICS |

(After Karwowski, 2005)

# Categories of Research Methods

- **Experimental**: controlled testing to examine cause-effect
- **Observational**: watching real-world behavior
- **Survey-based**: collecting subjective data
- **Mixed Methods**: combining qualitative and quantitative approaches

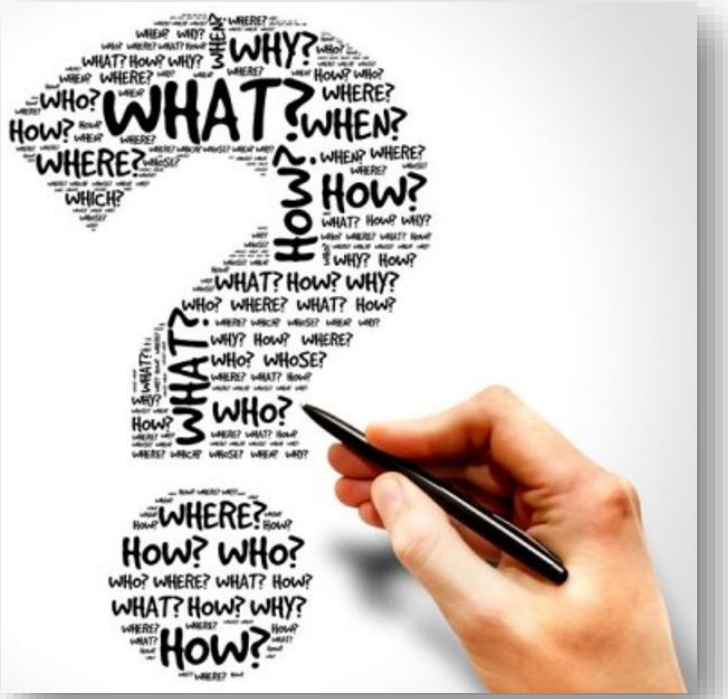
# Stages of research (Williges, 1995)



# Step 1: Define problem and hypotheses

- What is your **research question**?
  - e.g., “The purpose … was to investigate the effect of [touch screen] button and gap size on performance by individuals with varied motor abilities.” (Chen et al., 2013)
- What cause/effect relationships you expect to observe?

  **(Research hypotheses)**
  - e.g., Larger button size will increase accuracy and reduce task time
  - e.g., Greater gaps will increase accuracy and task time
- What are **potential variables** in the study?
  - e.g., Button size (measured in mm)
  - e.g., Gap size (between buttons; measured in mm)
  - e.g., Participant state: healthy and with disability (nominal classification)
  - e.g., Task time (duration; measured in sec.)
  - e.g., Percent error (select wrong button)
  - e.g., Percent miss (miss intended button target)



# Hypothesis Generation

- A good hypothesis clearly defines what variables are involved and what relationship you expect between them
- Define the variables involved and the relationship between them

- Precise: Clearly defines variables and expected effect
- Meaningful: Tied to a real-world problem or theoretical construct
- Testable: Can be studied using empirical methods
- Falsifiable: It’s possible for the data to disprove it

# Statistical hypothesis testing

- Null hypothesis ($H_0$): There is <u>no effect</u> of button size on touch screen user accuracy
- Alternative hypothesis ($H_A$): Button size affects accuracy

| | | If $H_0$ is... | |
|---|---|---|---|
| | | true. | false. |
| What you end up deciding to do with $H_0$ | reject | **Type I error, α (1 – specificity)** | **Correct (sensitivity)** |
| | fail to reject | **Correct (specificity)** | **Type II error, β (1 – sensitivity)** |

# Step 2: Experiment planning

- What are variables you want to manipulate?
- What are observations you want to make?

  How do we "operationalize" or find a way to measure the thing we are interested in?
- How many participants do you need?
- How will data be collected?
  - e.g., Do you want to take multiple measures on single subject at different points in time?

- **Independent variables (IV)** – Controlled manipulations of task, system or environment
  - Also called "Condition", "Factor", "predictor", "Indicator" or "Treatment"
  - Levels - Different settings of manipulation or condition
- **Dependent variable (DV)** – Response measures you intend to observe for effects of condition
  - Also called "Construct" or "Outcome measure"
  - Response to treatment (i.e. independent variable)
- How many people (**sample size**) do you want to recruit for study?
  - Depends on: desired effect size; significance criterion setting (tolerable $\alpha$); and statistical power (desired beta)

# What are the hypotheses, IVs, DMs…

1. The relationship between the amount of time couples spend together and their relationship satisfaction.

2. The relationship between number of hours studying and test performance.

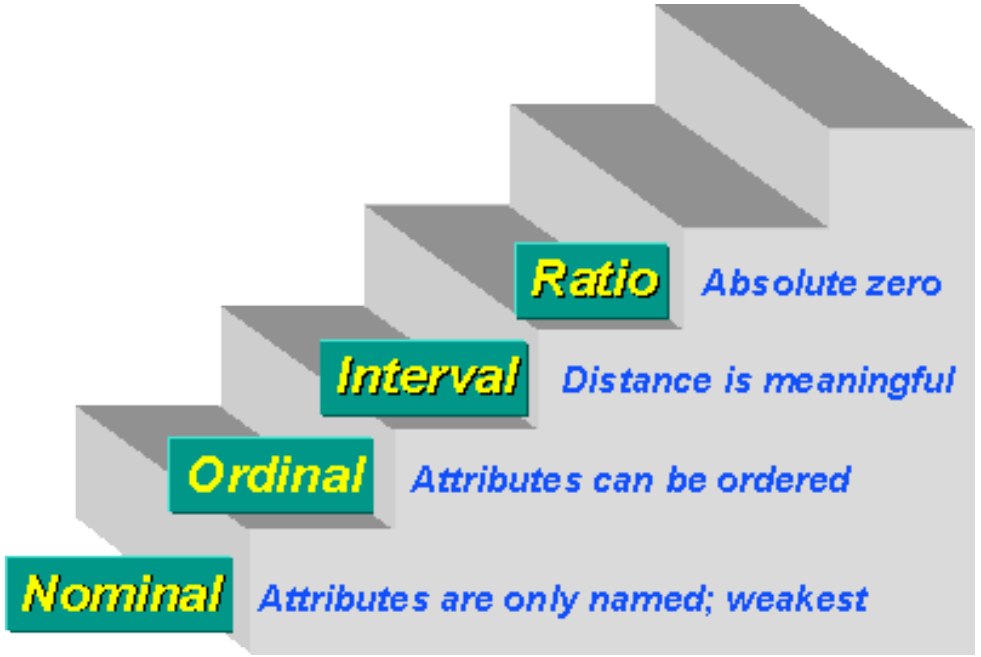
3. The relationship between the physical attractiveness of a political candidate and voters' opinions of him/her.

4. The relationship between time spent listening to Heavy Metal music and aggression.

# Levels of Measurement

**Table 2.** Scales of measurement. Measurement occurs when we use a schema (usually numbers) to represent empirical operations. The schemapiric outcome is called a scale. The four main scale types are shown below. Each type of scale uses different features of the number system because numbers can express identity, order, difference, or ratio. The key that defines the type of scale lies in the permissible number transformations, that is to say, the ways in which we can alter the numbers and still represent all the empirical information, with no loss. (Adapted from Stevens 1946.)

| Scale | Operations we perform | Permissible transformations | Some appropriate statistics | Examples |
|---|---|---|---|---|
| Nominal | Identify and classify | Substitution of any number for any other number | Number of cases<br>Mode<br>Contingency correlation | Numbering football players<br>Model numbers |
| Ordinal | Rank order | Any change that preserves order | Median<br>Percentiles<br>Rank-order correlation | Preference lists<br>Hardness of minerals<br>Rank lists |
| Interval | Find distances or differences | Multiplication by a constant<br>Addition of a constant | Mean<br>Standard deviation<br>Product-moment correlation | Temperature Fahrenheit<br>Temperature Celsius<br>Calendar time<br>Standard scores |
| Ratio | Find ratios, fractions, or multiples | Multiplication by a constant only | Geometric mean<br>Percent variability | Length, weight, numerosity, duration, and most physical scales<br>Temperature Kelvin<br>Loudness in sones |



# Levels of Measurement

| | Parametric tests | Nonparametric test | Shall we use a normality test? |
|---|---|---|---|
| **Large samples** (> 100) | **Robust.** P-value will be nearly correct, sometime even if population is fairly far from Gaussian population. | **Powerful.** If the population is Gaussian, the P-value will be nearly identical to the P-value we would have obtained from parametric test. With large sample sizes, nonparametric tests are almost as powerful as parametric tests. | **Useful.** We may use a normality test to determine whether the data are sampled from a Gaussian population. |
| **Small samples** (< 15) | **Not robust.** If the population is not Gaussian, the P-value may be misleading. | **Not powerful.** If the population is Gaussian, the P-value obtained from a nonparametric test may be higher than the P-value obtained from a parametric test. | **Not very useful.** It has little power to discriminate between Gaussian and non-Gaussian populations. |

# Example

**Fill-in "yes" or "no" for identified response characteristics:**

| Level of Measurement | Characteristics | | | Example |
|---|---|---|---|---|
| | Order | Equal Intervals | True Zero Point | |
| **Nominal** | NO | NO | NO | Gender (0-female; 1-male) |
| **Ordinal** | YES | NO | NO | Rank in class |
| **Interval** | YES | YES | NO | Relative temperature (°F/°C) |
| **Ratio** | YES | YES | YES | Speed, # of errors |

**Match example items below to classes of measurement:**

Gender (0-female; 1-male)

Rank in class

Speed, # of errors

Relative temperature (°F/°C)

# One IV vs. Multiple IVs

- One IV Example:
  - IV: Display brightness (low vs. high)
  - DV: Task completion time
- Multiple IVs Example:
  - IV1: Display brightness (low vs. high)
  - IV2: Font size (small vs. large)
  - DV: Task completion time
- Why multiple IVs? To explore interactions and real-world complexity

# Main Effects vs. Interactions

- Main Effect: Effect of one IV on the DM, averaging across other IVs
  - Think of it as asking: *"If we only look at this one factor, does it make a difference?"*
- Interaction Effect: Effect of one IV depends on the level of another IV
  - It answers: *"Does the impact of one factor change depending on the other factor?"*
- Suppose you run an experiment on **font size (small vs. large)** and **screen brightness (dim vs. bright)** to see how they affect **reaction time**.
  - A main effect of font size means that, on average, students read faster with large fonts compared to small fonts, regardless of screen brightness.
  - A main effect of brightness means that, on average, students read faster on bright screens compared to dim ones, regardless of font size.
  - Maybe large fonts only improve reaction time when the screen is dim but make no difference when the screen is bright.
  - That means the effect of font size depends on brightness → that's an interaction.

# Step 3: Conduct study

- Conduct pilot study to verify procedures and sample size
  - Follow experiment plan (design of experiment) for 2-3 subjects
  - Modify procedures to prevent errors and systematic variance (uncontrolled factor exposure)
  - Use pilot data (condition means and variances) to calculate sample size, given significance criterion and desired statistical power
- Conduct full experiment using refined procedure and verified sample size

# Step 4: Analyze the data

- Before doing any analyses, **look at descriptive statistics**:
  - Total number of participants (n); any missing data?
    - May dictate appropriate
  - Means, SD, histograms on DVs, trends of responses across levels of factors
  - Visualize the data (boxplots, stem-and-leaf plots, etc.)
  - **Why do this work?**
    - Can develop idea of what data is telling you before conducting inferential statistics (look at trends).
- Apply appropriate inferential statistics
  - ***t*-test** – simple single factor with two levels
  - **Simple linear regression** – single factor data model with continuous predictor
  - **Multiple regression** – multi-factor data model with continuous predictors
  - **Analysis of variance (ANOVA)** – single and multi-factor data models including interaction effects
    - Multivariate analyses – Analysis of effects on multiple responses accounting for potential correlations among responses
    - Repeated measures analysis – Multiple observations on same experimental unit accounting for potential dependence of observations, use linear mixed models

# Step 5: Draw conclusions

- Are your results generalizable?
  - Can they be extended to broader user population?
  - What type of validity is this?
    - External validity
- Always keep in mind statistical significance criterion ($p<0.05$)…
  - An arbitrary number
  - **Any design recommendations come with possibility of failure in application**
  - Would you consider values other than 0.05?
    If so, why?
    - Could consider $p<0.01$
    - Provide designers with greater assurance of effectiveness of design recommendation.

# Limitations in research

- Want "true" behavior of subjects we study:
  - Observation necessarily leads to changes in behavior
  - Persons under time study perform manual work ~12% faster (Frievalds, 2013)
  - Access to subjects may be limited in field research due to normal production operations
- Use of sample population:
  - Want to make inferences on population behavior but don't have resources to collect data on every person
  - Select individuals at random from population → sampling
  - Each sample yields different group of individuals and variance in observed mean response
    - Referred to as sampling error
    - Smaller sample size – greater sampling error
    - May need to conduct multiple experiments to estimate confidence interval on "true" mean
- Use of subjective response measures (e.g., surveys, ratings, interviews)
  - Influenced by recall of events
  - Influenced by performance bias (i.e., "workload" is often rated as low if task performance was "good")
  - Can have limited face validity due to contamination – subjects may consider task demands other than those of interest to analyst

# Challenges to HF/E research

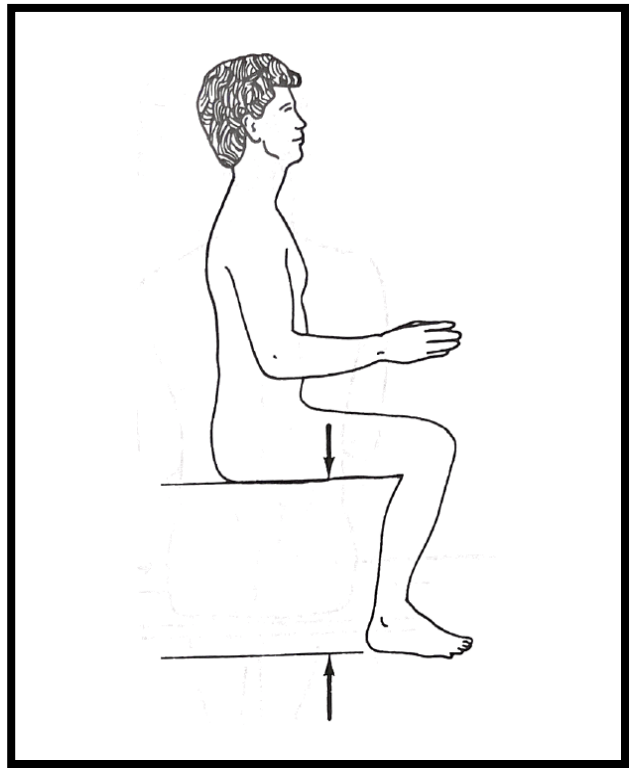
- **Limited funding**
  - May dictate sample size and sensitivity of analysis
- **Limited time**
  - May dictate number of factor levels that can be tested and understanding of relationship between predictors and response (actual trend)
- **Safety issues** (in testing)
  - Can lead to systematic error in responses (e.g., subject simulator sickness; subject perception of condition threat to well-being)
- What are **some possible alternatives to circumvent**?
  - Need to consider design options (within- vs. between-subjects).
  - Need to consider experimental procedure options (necessary duration of test trials to observe effect)

# What did we learn today?

- Other ways to describe research methods
  - Basic vs. Applied
  - Quantitative vs. Qualitative
  - Descriptive vs. Explanatory
  - Field vs. Laboratory
- Categories of research methods
  - Experimental
  - Observational
  - Survey-based
  - Mixed Methods
- Stages of research
  - Define (RQs, hypotheses, potential variables…)
  - Plan (IVs, DMs, sample size, experimental design, experimental procedure…)
  - Conduct (pilot study first…)
  - Analyze (t-test? ANOVA? …)
  - Interpret (validity… conclusions…)
- Challenges and limitations in HFE research

# Next Time!

- New module: Engineering Anthropometry!
- Application of anthropometric data
- Importance in Engineering Design
- Measurement Devices and Techniques
- Factors that affect anthropometric data
- Categories of Anthropometric Data
  - Dive into Static Anthropometry (Structural Dimensions)



**Popliteal Height**

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LEADING THE CHARGE, CHARGING AHEAD

# Appendix Slide – Hypothesis Generation Exercise

| Question | Hypothesis |
| --- | --- |
| Does a training program in driver safety result in a decline in accident rate? | People who take a driver safety course will have a lower accident rate than those who do not take the course. |
| Who are better in math, men or women? | Women are better at math than men. |
| What is the relationship between age and cell phone use? | Cell phone use is higher for younger adults than for older adults. |
| Is there a relationship between education and income? | Income increases with years of education. |
| Can public education reduce the occurrence of AIDS? | The number of AIDS cases is inversely related to the amount of public education about the disease. |

Source: http://psychology.ucdavis.edu/faculty_sites/sommerb/sommerdemo/stat_inf/null.htm